# Chapter 2 Answers

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## Lemma

We must show that, from $F_1$ and $F_2$, the following two conditions hold:
(1): We need to show that $\sim\exists x\phi(x)$ is true iff $\sim\phi(a)$ is true for every parameter $a$. Suppose that $\sim\exists x\phi(x)$ is true (under some valuation $v$). Thus, by $F_2$, the sentence $\phi(a)$ is not true under $v$ for any parameter $a$. Because $\phi(a)$ is not true under $v$ for any parameter $a$, it follows that $\sim\phi(a)$ is true under $v$ for every parameter $a$ (by $B_1$). On the other hand, suppose that $\sim\phi(a)$ is true under $v$ for every parameter $a$. Thus, by $B_1$, $\phi(a)$ is false under $v$ for every parameter $a$ (by $B_1$). Thus, by $F_2$, $\exists x\phi(x)$ is not true under $v$, meaning, by $B_1$, $\sim\exists x\phi(x)$ is true under $v$.
(2): We need to show that $\sim\forall x\phi(x)$ is true iff $\sim\phi(a)$ is true for at least one parameter $a$. On the one hand, suppose that $\sim\forall x\phi(x)$ is true under some valuation $v$. Thus $\forall x\phi(x)$ is not true under $v$ (by $B_1$), meaning, by $F_1$, $\phi(a)$ is not true for every parameter $a$. Thus, for the parameter $e$ for which $\phi(e)$ is not true, $\sim\phi(e)$ is true by $B_1$. On the other hand, suppose that $\sim\phi(a)$ is true for at least one parameter $a$, call it $e$. Thus, by $B_1$, $\phi(e)$ is false, meaning, by $F_1$, $\forall x\phi(x)$ is not true, and therefore, by $B_1$, $\sim\forall x\phi(x)$ is true.

## 1

We suppose that $v$ is a Boolean valuation such that the following two conditions hold (for all sentences $\gamma$ and $\delta$):
(1) if $\gamma$ is true under $v$, then for every parameter $a$, the sentence $\gamma(a)$ is true under $v$, and
(2) if $\delta$ is true under $v$, then $\delta(a)$ is true under $v$ for at least one parameter $a$.
We must prove that $v$ is a first-order valuation. Thus, we must derive $F_1'$ and $F_2'$. Because we have already have (1) and (2), we only need to prove the converses

of (1) and (2).
(1): In this case, we need to prove that, if for every parameter $a$, the sentence $\gamma(a)$ is true under $v$, then $\gamma$ is true under $v$.

Case 1: In this case, $\gamma$ is of the form $\forall x\phi(x)$. So suppose that $\phi(a)$ is true under $v$ for every paramter $a$. By $B_1$, it follows that $\sim\phi(a)$ is false under $v$ for every parameter $a$, meaning, by (2), that $\sim\forall x\phi(x)$ is not true, and is therefore false (because every sentence $X$ is either assigned the value $t$ or $f$), meaning, by $B_1$, $\forall x\phi(x)$ is true.

Case 2: In this case, $\gamma$ is of the form $\sim\exists x\phi(x)$. So suppose that $\sim\phi(a)$ is true for every parameter $a$, meaning, by $B_1$, that $\phi(a)$ is false for every parameter $a$. Thus, by (2), $\exists x\phi(x)$ is not true, and is therefore false (for the same reason as in Case 1), meaning, by $B_1$, that $\sim\exists x\phi(x)$ is true.

(2): In this case, we need to prove that, if $\delta(a)$ is true for at least one parameter $a$ under $v$, then $\delta$ is true under $v$.

Case 1: In this case, $\delta$ is of the form $\exists x\phi(x)$. So suppose that $\phi(a)$ is true for at least one parameter $a$, call it $e$. By $B_1$, $\sim\phi(e)$ is false, meaning, by (1), $\sim\exists x\phi(x)$ is not true, and is therefore false, and thus, by $B_1$, $\exists x\phi(x)$ is true.

Case 2: In this case, $\delta$ is of the form $\sim\forall x\phi(x)$. So suppose that $\sim\phi(a)$ is true for at least one parameter $a$, call it $e$. By $B_1$, $\phi(e)$ is false, meaning, by (1), $\forall x\phi(x)$ is not true, and therefore false, meaning, by $B_1$, $\sim\forall x\phi(x)$ is true.

## 2

We are to show that every Magic set is a truth-functional basis for propositional logic.

On the one hand, suppose that some pure sentence $X$ is valid. Thus, it is true under all first-order valuations. We must show that $X$ is a tautological consequence of some finite subset $M_0$ of $M$. That is, we must show that under every *Boolean* valuation that satisfies $M_0$, $X$ is true. Well, by $M_1$, every Boolean valuation that satisfies $S$ is also a first-order valuation. Thus, $X$ is true under all Boolean valuations that satisfy $M$, because $X$ was assumed to be true under all first-order valuations, and every Boolean valuation that satisfies $M$ is also a first-order valuation. Thus, $X$ is a tautological consequence of $M$. By the Corollary to the Denumerable Compactness Theorem for Propositional Logic, it is also a logical consequence of some finite subset $M_0$ of $M$.

On the other hand, suppose that some pure sentence $X$ is a truth-functional consequence of some finite subset $M_0$ of $M$. Thus, $M_0 \bigcup \sim X$ is not first-order satisfiable (every first-order valuation is also a Boolean valuation). This means that $\sim X$ is not first-order satisfiable. For, if it were, by $M_2$, $M_0 \bigcup \sim X$ would also be first-order satisfiable. Thus, $X$ is valid (true under every first-order valuation).

# 3

We must show that every complete regular set is also a magic set. Recall that a complete regular set $M$ is a set such that for every $\gamma$, $\gamma \supset \gamma(a) \in M$ for every parameter $a$, and for every $\delta$, $\delta \supset \delta(a) \in M$ for at least one parameter $a$, and that every finite subset of $M$ is a regular set. Because we are showing that every complete regular set is a magic set, it suffices to show that $M$ satisfies the conditions $M_1, M_2$.

$M_1$: We must show that every Boolean valuation that satisfies $M$ is also a first-order valuation. To that end, it suffices to show that:
If $\gamma$ is true, then $\gamma(a)$ is true for every parameter $a$,
If $\delta$ is true, then $\delta(a)$ is true for at least one parameter $a$.
(which implies a first-order valuation by Problem 1). So, suppose $\gamma$ is true. Thus, $\gamma \supset \gamma(a) \in M$ for every parameter $a$. Because this is a Boolean valuation, and $\gamma$ is true, $\gamma(a)$ is likewise true for $\gamma(a)$ for every parameter $a$ (by $B_4$). On the other hand, suppose that $\delta$ is true. Thus, $\delta \supset \delta(a) \in M$ for at least one parameter $a$. Again, because this is a Boolean valuation, and $\delta$ is true, $\delta(a)$ is likewise true for at least one parameter $a$ (again by $B_4$).

$M_2$: We must show that, for every finite set $S_0$ of pure sentences, and every finite subset $M_0$ of $M$, if $S_0$ is first-order satisfiable, so is $M_0 \bigcup S_0$. So, suppose that $S_0$ is a finite set of pure sentences, and that it is first-order satisfiable. Now consider any finite subset $M_0$ of $M$. By the definition of a complete regular set, it is a regular set. Thus, it can be split into a sequence

$$Q_1 \supset Q_1(a_1), ..., Q_n \supset Q_n(a_n)$$

where, if $Q_i$ is a $\gamma$, $a_i$ is any parameter $a$, and if $Q_i$ is a $\delta$, then $a_i$ is a parameter $a$ such that $a$ does not occur in $\delta$ or in any previous $Q$'s. So, what we really need to show is that, if a set $S$ is first-order satisfiable, $S \bigcup \gamma \supset \gamma(a)$ is first-order satisfiable for every parameter $a$, and $S \bigcup \delta \supset \delta(a)$ is first-order satisfiable, provided $a$ does not occur in $\delta$ or $S$. This would let us show that

$$S_0 \bigcup Q_1 \supset Q_1(a_1) \bigcup ... \bigcup Q_n \supset Q_n(a_n)$$

is first-order satisfiable, which is the same as saying $S_0 \bigcup M_0$ is first-order satisfiable. There are then two cases: $Q_i$ is a $\gamma$, or is a $\delta$.

$\gamma$: in this case, $Q_i$ is a $\gamma$, meaning $Q_i \supset Q_i(a_i)$ is of the form $\gamma \supset \gamma(a_i)$. Suppose $S$ is first-order satisfiable under some interpretation $I$. Well, $\gamma \supset \gamma(a_i)$ is satsified by *every* first-order valuation (which can easily be shown, if one desires), meaning $S \bigcup \gamma \supset \gamma(a_i)$ is obviously also first-order satisfiable.

$\delta$: in this case, $Q_i$ is a $\delta$, meaning $Q_i \supset Q_i(a_i)$ is of the form $\delta \supset \delta(a_i)$. Suppose $S$ is first-order satisfiable under some interpreation $I$. If $\delta$ is false under $I$, then $\delta \supset \delta(a_i)$ is true under $I$. If $\delta$ is true under $I$, then $S \bigcup \delta$ is satisfiable, meaning $S \bigcup \delta \supset \delta(a_i)$ is satisfiable by Problem 2 of Chapter 9 of the Beginners Guide[Smullyan, 2014] (because we assumed that $a$ does not occur in $\delta$ or any of the previous $Q_i's$, and that $S_0$ was a set of pure sentences, meaning $a$ of course does not occur in them). In either case, $S \bigcup \delta \supset \delta(a_i)$ is satsifiable.

# 4

We are to find a quicker proof of the Regularity theorem, using Magic sets. It has been shown that a complete regular set exists, and by Problem 3, it is a Magic set. By Problem 2, it is therefore a truth-functional basis for first-order logic. So, suppose that $X$ is a valid pure sentence. Thus, $X$ is truth-functionally implied by some finite subset $M_0$ of $M$. By the definition of a complete regular set, $M_0$ is a regular set. Thus, $X$ is tautologically implied by some regular set, which is what we were required to prove.

# System $G_0$

## Correctness of $G_0$

The Axiom of the system is obviously correct. For, whenever the set of formulas $U, X$ is true, $X$ is true, meaning it is obviously true that *at least one* of the formulas $V, X$ are true for any set of formulas $V$.

$C_1$: This is obvious, as $X, Y$ are defined to both be true whenever $X \wedge Y$ is true.

$C_2$: If whenever all elements of $U$ are true, at least one element of $V, X$ and $V, Y$ is true, on the one hand, if $V$ is the element which is true, it follows that at least one element ($V$) of $V, X \wedge Y$ is true. On the other hand, if $V$ is false, it would then follow that both $X, Y$ are true, therefore meaning $X \wedge Y$ is the element which is true.

$D_1$: If whenever all elements of $U$ are true then at least one element of $V, X, Y$ is true, if $V$ is the element which is true, it follows that at least one element of $(V)$ of $V, X \vee Y$ is true. On the other hand, if $V$ is false, it would then follow that either $X$ or $Y$ is true, therefore meaning at least one element of $V, X \vee Y$ is true.

$D_2$: We are given that at least one element of $V$ is true whenever both $U, X$ are true, or when both $U, Y$ are true. By definition of $X \vee Y$, it follows that all elements of $U, X \vee Y$ are true iff either $X$ or $Y$ is true, which would imply that at least one element of $V$ is true.

$I_1$: We are given that at least one element of $V, Y$ is true whenever all elements of $U, X$ are true. Thus, if all elements of $U$ are true, either $X$ is true or false. If it is false, at least one element $(X \supset Y)$ of $V, X \supset Y$ is true, while if it is true, it would follow by the assumptions that either $V$ or $Y$ is true. If it is $V$ that is true, this is the element that is true. If it is $Y$ that is true, then $X \supset Y$ is also true.

$I_2$: We are given that at least one element of $V, X$ is true whenever $U$ is true, and that at least one element of $V$ is true whenever all elements of $U, Y$ are true. Thus, if all elements of $U, X \supset Y$ are true, it follows that either $V, X$ are true. If it is $V$ that is true, we are done. If it is $X$ that is true, we know that $Y$ is also true, meaning $V$ would also be true by the assumptions.

$N_1$: We are given that at least one element of $V$ is true whenever all elements of $U, X$ are true. If $U$ is true, either $X$ is true or false: if it is true, the assumptions prove that $V$ is also true. If it is false, then $\sim X$ would be true by definition.

$N_2$: We are given that when all elements of $U$ are true, at least one element of $V, X$ is true. Thus, if all elements of $U, \sim X$ are true, $V$ would be the true element implied by $U$, as $X$ is false.

# Problem 5

(a): On the one hand, suppose that the sequent $U, X \rightarrow V$ is true under some valuation $v$. There are two cases: either at least one of $U, X$ are false under $v$, or both of $U, X$ are true under $v$. In the first case, if it is $U$ that is false, $U \rightarrow V, \sim X$ is automatically true. Otherwise, if $X$ is false, $\sim X$ is true, meaning $U \rightarrow V, \sim X$ is also true. On the other hand, if both of $U, X$ are true under $v$, $V$ is also true under $v$, therefore meaning that $U \rightarrow V, \sim X$ is also true under $v$.

On the other hand, suppose that $U \rightarrow V, \sim X$ is true under some valu-

ation $v$. If $U$ is false, we are done, as this would trivially mean that $U, X \to V$ is true. On the other hand, if $U$ is true, either $V$ or $\sim X$ is true as well. If it is $V$, we are done, as $U, X \to V$ would be true. If it is $\sim X$, then $X$ would be false, therefore meaning $U, X \to V$ is trivially true.

(b): On the one hand, suppose that $U \to V, X$ is true. If $U$ is true, either $V$ or $X$ is true. If it is $X$ that is true, then $U, \sim X \to V$ is trivially true. If it is $V$ that is true, then $U, \sim X \to V$ is of course also true. If $U$ is false under $v$, then $U, \sim X \to V$ is trivially true under $v$.

On the other hand, suppose that $U, \sim X \to V$ is true under $v$. If both of $U, \sim X$ are true, then $X$ is false, therefore meaning $U \to V, X$ is trivially true. On the other hand, either of $U, \sim X$ would be false. If it is $U$ that it false, $U \to X, V$ is trivially true. If it is $\sim X$ that is false, $X$ is true, therefore meaning $U \to V, X$ is also true under $v$.

# 6

As for the axiom, suppose that $|S, TX, FX|$ is provable, where $S$ is a set of signed formulas. Let $TX_1, ..., TX_n$ be all of the signed formulas with a $T$ in $S$, and $FY_1, ..., FY_k$ be all of the signed formulas with an $F$ in $S$. Thus $|S, TX, FX|$ is just $X_1, ..., X_n, X \to X, Y_1, ..., Y_k$, which is just the same thing as $U, X \to V, X$ holding, as we could just let $X_1, ..., X_n$ be the set $U$, and $Y_1, ..., Y_k$ be the set $V$. On the other hand, suppose that we take the axiom of the system $G_0$. Thus, for any sets $U, V$ of formulas, and any formula $X$, $U, X \to X, V$ is provable. Again, consider some set $S$ of signed formulas, and suppose that $TX_1, ..., TX_n$ be all of the signed formulas with a $T$ in $S$, and $FY_1, ..., FY_k$ be all of the signed formulas with an $F$ in $S$. Thus $|S, TX, FX|$ is just $X_1, ..., X_n, X \to X, Y_1, ..., Y_k$. Because the axiom of $G_0$ holds for any sets $U$ and $V$, we can simply take $U$ to be the set $X_1, ..., X_n$, and $V$ to be the set $Y_1, ..., Y_k$, thus proving that $|S, TX, FX|$ is provable.

We now need to show that the inferences rules of $G_0$ in uniform notation are equivalent to those inferences rules of $G_0$ not in uniform notation. First, let us show that the rules in uniform notation imply those rules not in uniform notation.

$C_1$: Consider any two sets of formulas $U$ and $V$. We can take $S$ to be the set of formulas $TX_1, ..., TX_n, FY_1, ..., FY_k$ such that $X_1, ..., X_n = U$ and $Y_1, ..., Y_k = V$ (for making the following proofs shorter, we will call the set $S$ constructed in this way the *signed set for* $U, V$. By $A$, if $|S, TX, TY|$ is provable, that is, $U, X, Y \to V$, so is $|S, TX \wedge Y|$, that is, $U, X \wedge Y \to V$.

$C_2$: Consider the signed set $S$ for any two sets $U$ and $V$. By $B$, if $|S, FX|$ is provable, that is, $U \rightarrow V, X$, and $|S, FY|$ is provable, that is, $U \rightarrow V, Y$, then $|S, FX \wedge Y|$ is provable, that is, $U \rightarrow V, X \wedge Y$.

$D_1$: Consider the signed set $S$ for $U$ and $V$. By $A$, if $|S, FX, FY|$ is provable, that is, $U \rightarrow V, X, Y$, then $|S, FX \vee Y|$ is also provable, that is, $U \rightarrow V, X \vee Y$.

$D_2$: Consider the signed set $S$ for $U$ and $V$. By $B$, if $|S, TX|$ is provable, that is, $U, X \rightarrow V$, and $|S, TY|$, that is, $U, Y \rightarrow V$, is provable, then so is $|S, TX \vee Y|$, that is, $U, X \vee Y \rightarrow V$.

$I_1$: Consider the signed set $S$ for $U$ and $V$. By $A$, if $|S, TX, FY|$, that is, $U, X \rightarrow V, Y$, is provable, so is $|S, FX \supset Y|$, that is, $U \rightarrow V, X \supset Y$.

$I_2$: Consider the signed set $S$ for $U$ and $V$. By $B$, if $|S, FX|$, that is $U \rightarrow V, X$ is provable, and $|S, TY|$, that is $U, Y \rightarrow V$, is provable, so is $|S, TX \supset Y|$, that is $U, X \supset Y \rightarrow V$.

$N_1$: Consider the signed set $S$ for $U$ and $V$. By $A$, if $|S, TX|$, that is $U, X \rightarrow V$, is provable, so is $|S, F{\sim}X|$, that is $U \rightarrow V, {\sim}X$.

$N_2$: Consider the signed set $S$ for $U$ and $V$. By $A$, if $|S, FX|$, that is $U \rightarrow V, X$, is provable, so is $|S, T{\sim}X|$, that is $U, {\sim}X \rightarrow V$.

We now turn to showing that the inferences rules of $G_0$ imply $A$ and $B$.

$A$: If $\alpha$ is $TX \wedge Y$, this can be shown using $C_1$. If $\alpha$ is $FX \vee Y$, this can be shown using $D_1$. If $\alpha$ is $FX \supset Y$, this can be shown using $I_1$. If $\alpha$ is $T{\sim}X$, this can be shown using $N_2$. If $\alpha$ is $F{\sim}X$, this can be shown using $N_1$.

$B$: If $\beta$ is $TX \vee Y$, this can be shown using $D_2$. If $\beta$ is $FX \wedge Y$, this can be shown using $C_2$. If $\beta$ is $TX \supset Y$, this can be shown using $I_2$. The cases where $\beta$ is $T{\sim}X$ or $F{\sim}X$ are the same as in the $\alpha$ case.

## A set $S$ is unsatisfiable iff its corresponding sequent $|S|$ is a tautology

On the one hand, suppose that a non-empty set $S$ of signed formulas is unsatisfiable. There are then multiple cases. In the first case, suppose that there are no formulas of the form $TX$. Thus, $|S|$ is of the form $\rightarrow Y_1, ..., Y_k$

where every $Y_i$ is the formula $FY_i$ without the sign $F$. We must show that at least one of $Y_1, ..., Y_k$ is true. We are given that $FY_1, ..., FY_k$ is unsatisfiable, meaning, under every interpretation $I$, there is some formula $FY_j$ that is false. Under this same interpretation $I$, it is obvious that $Y_j$ (removing the sign) is true, thus meaning at least one of $FY_1, ..., FY_k$ is true. In the second case, suppose that the set $S$ contains no formulas of the form $FY$. Thus $|S|$ is of the form $X_1, ...X_n \rightarrow$. We need to show that at least one of $TX_1, ..., TX_n$ is false. But this is immediate, as it was assumed that one of $TX_1, ..., TX_n$ was false under every interpretation $I$, which is the same as saying one of $X_1, ..., X_n$ is false under every interpretation $I$. In the last case, the set $S$ contains at least one formula of the form $TX$ and at least one of the form $FY$. Thus $|S|$ is of the form $X_1, ..., X_n \rightarrow Y_1, ..., Y_k$. If every element of $X_1, ..., X_n$ is true, this would mean, by the assumption that $S$ is not simultaneously satisfiable, that at least one of $FY_1, ..., FY_k$ is false (call this element $FY_K$). As before, it can be easily seen that $Y_k$ would then be true under this same interpretation, therefore meaning at least one of $Y_1, ..., Y_k$ is true.

On the other hand, suppose that $|S|$ is a tautology for some non-empty set of formulas $S$. There are then 3 cases:

Case 1: In this case, $S$ does not contain any formulas of the form $TX$. We know that $\rightarrow Y_1, ..., Y_k$ is a tautology, meaning under every interpretation, at least one of $Y_1, ..., Y_k$, which is the same as saying, under every interpreation, at least one of $FY_1, ..., FY_k$ is false, thus meaning $S$ is unsatisfiable.

Case 2: In this case, $S$ does not contain any formulas of the form $FY$. We know that $X_1, ..., X_n \rightarrow$ is a tautology, meaning, under every interpretation, at least one of $X_1, ..., X_n$ are false, which is the same as saying at least one of $TX_1, ..., TX_n$ are false, which means that $S$ is unsatisfiable.

Case 3: In this case, $S$ contains at least one formula of the form $FY$, and at least one formula of the form $TX$. Thus we know that $X_1, ..., X_n \rightarrow Y_1, ..., Y_K$ is a tautology. There are then two cases. In the first case, at least one of $X_1, ..., X_n$ are false under an interpretation $I$, which is the same as saying at least one of $TX_1, ..., TX_n$ is false under $I$, which would mean $S$ is not simultaneously true under $I$. In the other case, all of $X_1, ..., X_n$ are true, which would mean that at least one of $Y_1, ..., Y_k$ are true, which is the same as saying that at least one of $FY_1, ..., FY_k$ is false, therefore meaning that $S$ is also not simultaneously satisfiable under this interpretation.

## First-order logic case

The case where we wish to prove that a set of first order formulas $S$ if unsatisfiable iff its correponding sequent is *valid* is identical to the above proof,

except replacing "tautology" with "valid".

# 7

We first show that the altered tableau system is correct, that is, if a set of formulas $S$ is satisfiable, then no altered tableau for $S$ can close. It sufficies to show that, if $S$ is satisfiable, then:
(1) If $A \in S$, then $S \bigcup A_1$ and $S \bigcup A_2$ are both satisfiable,
(2) If $B \in S$, then either $S \bigcup B_1$ or $S \bigcup B_2$ is satisfiable,
(3) If $C \in S$, then $S \bigcup C(a)$ is satisfiable for every parameter $a$,
(4) If $D \in S$, then $S \bigcup D(a)$ is satisfiable for every parameter $a$ new to $S$.

These can all be proven by replacing $\alpha, \beta, \gamma, \delta$ by $A, B, C, D$, respectively, in Chapter 9 of the Beginners Guide [Smullyan, 2014]. It is obvious that the use of an altered tableau rule is in effect using one of the four methods of expansion in the above description. Thus, if using those methods preserves satisfiability, then an altered tableau for a set of satisfiable formulas $S$ cannot close.

Next, we show that the altered tableau system is complete, that is, if an altered tableau for a set formulas $S$ is open, then $S$ is satisfiable. Consider a $S$ set of the following form:
(1) If $A \in S$, then $A_1 \in S$ and $A_2 \in S$,
(2) If $B \in S$, then either $B_1 \in S$ or $B_2 \in S$,
(3) If $C \in S$, then $C(a) \in S$ for every parameter $a$,
(4) If $D \in S$, then $D(a) \in S$ for at least one parameter $a$.
(5) There is no formula $X$ such that: $T\ X \in S$ and $T \sim X \in S$; or $T\ X \in S$ and $F\ X \in S$; or $F\ X \in S$ and $F \sim X \in S$.
Now, it is clear that if an altered tableau does not close, and runs on infinitely (i.e, is open), every branch on the altered tableau is such a set. Thus, if we can show that every set of the above form is satisfiable, we can show that, if an altered tableau for a set of formulas $S$ is open, then the set $S$ is simultaneously satisfiable (which would show by contraposition that the altered tableau system is complete). We can prove this by complete induction, that is, we must prove that there is some interpretation $I$ under which all elements of $S$ are true.

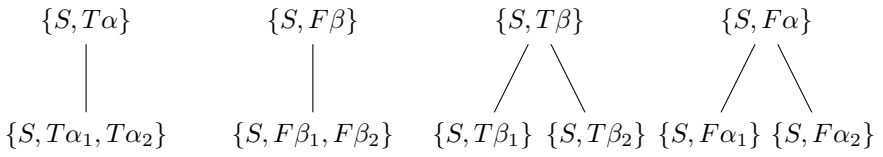
Base case: We must show that all formulas of degree 0 (atomic formulas) are satisfiable under some interpretation $I$ (and then show all the formulas of greater degrees are true under that same interpretation). Now, we can assign the value $T$ to any of the following formulas $X$: if $T\ X \in S$ or $F \sim X \in S$, then we assign $X$ the value of truth (and hence both $T\ X$ and $F \sim X$ would be true as well). Otherwise, if $F\ X \in S$ or $T \sim X \in S$, then we assign $X$ the value

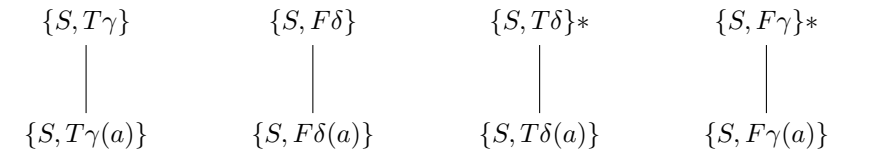
of falsity (and hence both $F\ X$ and $T\ {\sim}X$ would be true as well). If none of $T\ X, F\ {\sim}X, F\ X, T\ {\sim}X$ are in $S$, then it doesn't matter what value we assign to $X$. In this way, we ensure that all atomic formulas in $S$ are true under $I$.

Inductive case: Now, we assume that all formulas in $S$ of a degree less than $n$ are true under $I$. We must show that all formulas $X$ of degree $n+1$ in $S$ are also true under $I$. Suppose $X$ is of type $A$. Thus $A_1 \in S$ and $A_2 \in S$, which are thus both true under $I$, and hence $A$ is also true under $I$. Next, suppose $X$ is of type $B$. Thus either $B_1 \in S$ or $B_2 \in S$, and whichever one is in $S$, it is thus true under $I$, hence meaning $B$ is also true under $I$. Next, suppose that $X$ is of type $C$. Thus $C(a) \in S$ for every parameter $a$, which are all true under $I$, thus meaning $C$ is also true under $I$. In the last case, suppose $X$ is of type $D$. Thus $D(a) \in S$ for at least one parameter $a$, and all the formulas of the form $D(a) \in S$ are true under $I$ by the assumption, thus meaning $D$ is also true under $I$. Hence, the proof is complete.

So, we have shown the above type of set to be satisfiable. This means that, because every open altered tableau is such a set, any set of formulas $S$ which has an open altered tableau is satisfiable, and hence, by contraposition, the completeness of the altered tableau system has been proven.

Lastly, we must show that the system $GG$ is complete, that is, we must show that for any sequent which is valid, $GG$ provides a proof of that sequent. Now, let us transform any given sequent $X_1, ..., X_n \rightarrow Y_1, ..., Y_k$ to the set of formulas $\{T\ X_1, ..., T\ X_n, F\ Y_1, ..., F\ Y_k\}$. Now, if this set of formulas is unsatisfiable (which it must be if a sequent is valid), there is a proof of it using the altered tableau system (as the system was proven complete). So consider the proof of this set using the altered tableau system. First, let us convert the inference rules of the altered tableau method into their counterparts in uniform notation (which, in this case, we make to be a *block* tableau system, not an analytic one):



$*$Providing $a$ is not in the premise.

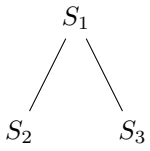
Now, for any of the above block tableau rules of the form:




we mean

$$\frac{|S_2|}{|S_1|}$$

by the *counterpart* of the rule. If the block rule is of the form:




we mean

$$\frac{|S_2|, |S_3|}{|S_1|}$$

by its counterpart. In fact, the counterparts of the eight block tableau rules listed above are the eight inference rules in the system $GG$ (as the reader can easily verify). Now let us suppose we have some given closed altered block tableau $\tau$ for a set of formulas $S$. By virtue of it being closed, all branches of $\tau$ will end with a block of one of the three forms:

$$\{S, TX, FX\}, \{S, TX, T{\sim}X\}, \{S, FX, F{\sim}X\}$$

Here is how we convert $\tau$ into a proof of $S$ in the system $GG$. First, if $\tau$ has $n$ closed branches, we place the corresponding sequents of all of the $n$ blocks at the bottom of each of those closed branches as the first $n$ steps of the proof. Because each of these blocks $S$ is of one of the above three forms, we know that its corresponding sequent $|S|$ is of one of the three forms:

$$U, X \to V, X;\ U, X, \sim X \to V;\ U \to V, X, \sim X$$

which are all axioms of the system $GG$. Hence, we are allowed to start the proof with these sequents. Next, we simply flip the altered block tableau upside down. This is doable as, in flipping it, all the block tableau rules are switched to become their counterparts (which we have shown to be the inference rules of $GG$).

## Exercise

Axiom 1: This is the sequent $U, X \to V, X$. Here, an interpolant is $X$. For, $U, X \to X$ is obviously valid, and $X \to V, X$ is also valid (and $V, X$ and $U, X$ clearly contain all parameters and predicates in $X$).

Axiom 2: This is the sequent $U, X, \sim X \to V$. Here, an interpolant is $f$. $U, X, \sim X \to f$ and $f \to V$ are obviously valid, and all the predicates and parameters of $f$ are in $V$ and $U, X, \sim X$ (as $f$ has no parameters or predicates).

Axiom 3: This is the sequent $U \to V, X, \sim X$. Here, an interpolant is $t$. $U \to t$ and $t \to V, X \sim X$ are both obviously valid (as $X \vee \sim X$ is valid), and, as in the previous axiom, $t$ has no parameters or predicates.

Rule $A_1$: Suppose that there is an interpolant $X$ for the sequent $U, \alpha_1, \alpha_2 \to V$. Then $X$ is obviously an interpolant for $U, \alpha \to V$, as whenever all of $U, \alpha$ are true, so would all of $U, \alpha_1, \alpha_2$ (and hence $X$ would also be true). Now, if $\alpha_1, \alpha_2$ contains any of the parameters or predicates of $X$, then so would $\alpha$, which takes care of the predicates and parameters of $X$ (as $V$ contains all of the parameters and predicates of $X$ by the assumption).

Rule $A_2$: Suppose that there is an interpolant $X$ for the sequent $U \to V, \beta_1, \beta_2$. As in the above rule, $X$ is also an interpolant for $U \to V, \beta$. $X \to V, \beta$ is valid, as whenever $X$ is true, either $V, \beta_1, \beta_2$ is true, and if it is $\beta_1$ or $\beta_2$, then $\beta$ would be true as well. The predicates and parameters also match up, as all predicates and parameters in $\beta_1, \beta_2$, are also in $\beta$.

Rule $B_1$: Suppose that there is an interpolant $X$ for the sequent $U, \beta_1 \to V$ and an interpolant $Y$ for the sequent $U, \beta_2 \to V$. Then $X \vee Y$ is an interpolant for the sequent $U, \beta \to V$. If all elements of $U, \beta$ are true, then either $X$ or $Y$

is true (as either $\beta_1, \beta_2$ would be true), meaning $X \vee Y$ is true. If $X \vee Y$ is true, then $V$ is true by the assumption of interpolancy for whichever of $X$ or $Y$ is true. As for the predicates and parameters of $X \vee Y$, it was assumed that $V$ contains the predicates and parameters of both $X$ and $Y$, while $U, \beta$ contains all the predicates and parameters of $X \vee Y$ (as $U, \beta_1$ and $U, \beta_2$ contain all the predicates and parameters of $X$ and $Y$, respectively, and $\beta$ contains all the predicates and parameters of $\beta_1$ and $\beta_2$).

Rule $B_2$: Suppose that there is an interpolant $X$ for the sequent $U \rightarrow V, \alpha_1$ and an interpolant $Y$ for the sequent $U \rightarrow V, \alpha_2$. Then $X \wedge Y$ is an interpolant for $U \rightarrow V, \alpha$. For, if all elements of $U$ is true, then so are both $X$ and $Y$ (that is, $X \wedge Y$) by the assumption. And, if $X \wedge Y$ is true, then at least one element of both $V, \alpha_1$ and $V, \alpha_2$ is true. If it is both $\alpha_1, \alpha_2$ which are true, then $\alpha$ is true as well. As for the predicates and parameters, $U$ contains the predicates and parameters of both $X$ and $Y$ by the assumption, and $V, \alpha$ contains all the predicates and parameters of $X \wedge Y$ as $\alpha$ contains all the predicates and parameters of $\alpha_1, \alpha_2$.

## Completion of the Proof of Craig's Lemma for Rule's C and D

Rule D: We need to show that, if $X$ is an interpolant for $U \rightarrow V, \gamma(a)$ and $a$ does not occur in $U$ or in $V, \gamma$, then $X$ is an interpolant for $U \rightarrow V, \gamma$. By the assumption of interpolancy, all parameters of $X$ occur in $V$, as well as $U, \gamma(a)$. Thus, because $a$ does not occur in $V$ (by the hypothesis), $a$ must also not occur in $X$. Now, it is clear $\gamma$ contains all parameters $b$ in $\gamma(a)$ such that $b \neq a$. And hence, for whatever parameters in $X$ that $\gamma(a)$ has, $\gamma$ also contains them (because these parameters must be $\neq a$). Thus, $V, \gamma$ contains all the parameters of $X$. Both $U$ and $V, \gamma$ also contain all the predicates of $X$, as the predicates in $\gamma$ are the same as those in $\gamma(a)$.

Now, $U \rightarrow X$ is valid by the assumption, and, because $X \rightarrow V, \gamma(a)$ was assumed valid, so is $X \rightarrow V, \gamma$ by rule D, thus proving $X$ is an interpolant for $U \rightarrow V, \gamma$.

Rule C: We need to show that, if $X$ is an interpolant for $U \rightarrow V, \delta(a)$, then if $a$ does not occur in $X$, or $a$ does occur in $V, \delta$, then $X$ is an interpolant for $U \rightarrow V, \delta$. If $a$ does occur in $X$ but not in $V, \delta$, then an interpolant for $U \rightarrow V, \delta$ is $\exists x \phi(x)$ where $\phi(x)$ is the result of replacing every occurence of $a$ in $X$ by a variable $x$ that does not occur in $X$. The first part of the assumption is obvious. Next, we must show the second part. Hence, assume that $X$ is an interpolant for $U \rightarrow V, \delta(a)$, and $a$ occurs in $X$ but not in $V, \delta$. We must show

that $\exists x\phi(x)$ is an interpolant for $U \to V, \delta$ where $\phi(x)$ is the result of replacing every occurence of $a$ in $X$ by a variable $x$ that does not occur in $X$.

First, note that $\phi(a) = X$. Now, let $\delta' = \exists x\phi(x)$. Hence $\delta'(a) = \phi(a) = X$, meaning $\delta'(a) \to V, \delta(a)$ (by the assumption) and thus $\delta'(a) \to V, \delta$ (by rule C). Thus $\delta' \to V, \delta$ (by rule D, and the fact that $a$ is not in $\delta, V$ or $\delta$). Also, because $U \to \delta'(a)$ is valid (by the assumption), so is $U \to \delta'$ (by Rule C). Thus, both $\delta' \to V, \delta$ and $U \to \delta'(a)$ are valid, and all predicates and parameters in $\delta'$ occur in both $U$ (by the assumption) and in $V, \delta$ (because $\delta'$ does not contain $a$, and $V, \delta(a)$ was assumed to have all parameters of $\delta'(a)$, and $\delta$ and $\delta'$ have all parameters of $\delta(a)$ and $\delta'(a)$, respectively, except $a$).

## Notes on Beth's definability Theorem

An important fact in proving the theorem is that changing around all occurences of a given predicate in a valid formula preserves validity. For example, the formula:

$$\forall x\,(Px \vee \sim Px)$$

is valid, meaning the formula:

$$\forall x\,(Rx \vee \sim Rx)$$

is valid for any other predicate $R$. To illustrate the more general fact, consider how first-order tableaus are completed. They have no dependency on the actual predicates (because we are searching for validity, and hence the predicates cannot take on a specific meaning), and only change depending on the logical structure of the formula, so substituting new predicates for all occurences of a given predicate will not change validity (as long as the new predicates we are substituting do not already occur in the formula).

The above fact is somewhat related to the following fact that is used in the proof. Suppose that we have:

$$(1)\ X \supset (Pa \equiv \phi\,(a))$$

being a valid formula. Then

$$X \supset \forall x\,(Px \equiv \phi\,(x))$$

is also valid, provided that $a$ does not occur in $X$ or

$$\forall x\,(Px \equiv \phi\,(x))$$

To see this, again note that, because we are testing validity, we are not considering only one domain. Hence, we take no notice of the values of predicates

in a certain domain, but again, only the formulas logical strucutre. What this means is that the above formula can only be valid if it does not lose its validity by switching around the parameters (in a similar fashion to switching around the predicates), and hence we can quantify it over all the variables in any domain. The only problem that could occur with this is if $a$ is in:

$$X \supset \forall x \left(Px \equiv \phi\left(x\right)\right)$$

This may be a problem if the validity of (1) relies upon $Pa$ or $\phi(a)$ making some connection with some other sub-formula with $a$ in it, in which case case we could not quantify the formula to be universal (as the connection between $Pa$ or $\phi(a)$ and that other sub-formula with $a$ would no longer occur, as the parameters would differ in the universal quantification). This fact can be generalized to the rule:

$$\frac{X \supset \left(Y \wedge \gamma\left(a\right)\right)}{X \supset \left(Y \wedge \gamma\right) *}$$

$*$Providing $a$ is not in the conclusion.

In the proof given on 48, note that $\phi\left(x\right)$ does not contain $a$, which allows us to receive the formula

$$X \supset \forall x \left(Px \equiv \phi\left(x\right)\right)$$

This is because $\phi(x)$ was defined to be the result of substituting $x$ for all occurences of $a$ in $Z$. But, because $Z$ was shown to contain no parameters other than $a$, it follows that $\phi(x)$ contains no parameters.

A similar fact can be observed for existential quantification:

$$\frac{\left(X \wedge \delta\left(a\right)\right) \supset Y}{\left(X \wedge \delta\right) \supset Y *}$$

$*$Providing $a$ is not in the conclusion.

This can be seen in a similar fashion: the validity of the formula does not depend on the specific value of the parameter, but only on the *strcuture* of the parameters in the formula.

# 8

Recall that $P'$ is a predicate distinct from all of $P, P_1, ..., P_n$ and which does not occur in any element of $S$ ($S$ is assumed to have no parameters). Recall also that $S'$ is the result of substituting $P'$ for $P$ in all occurences of $P$ in elements of $S$. Now, we assume that $P$ is explicitly definable from $P_1, ..., P_n$ with respect to $S$. We must show that it is implicitly so definable. Well, by the assumption

$$S \vdash \forall x \left(Px \equiv \phi\left(x\right)\right)$$

for some formula $\phi(x)$ such that its predicates are among $P_1, ..., P_n$. This is the same as saying the formula

$$S \supset \forall x \left(Px \equiv \phi\left(x\right)\right)$$

is valid, which implies the formula

$$S' \supset \forall x \left(P'x \equiv \phi\left(x\right)\right)$$

is valid as $\phi(x)$ does not contain the predicate $P$ (see the notes in the previous section about switching around predicates in valid formulas for an explanation). Thus

$$S \wedge S' \supset \forall x \left(P'x \equiv \phi\left(x\right)\right) \wedge \forall x \left(Px \equiv \phi\left(x\right)\right)$$

is valid, which implies:

$$S \wedge S' \supset \forall x \left(Px \equiv P'x\right)$$

which is to say:

$$S \bigcup S' \vdash \forall x \left(Px \equiv P'x\right)$$

which is what we were required to prove.

# 9

(a): First, suppose that the Regularity Theorem holds; hence every valid sentence $X$ is tautologically implied by some regular set $R$ such that no critical parameter of $R$ occurs in $X$. We must show that every unsatisfiable set has an assosciate. Well, suppose that we have some unsatisfiable set $S$. Now, let the elements of $S$ be $X_1, X_2, ..., X_n$. Thus $X_1 \wedge X_2 \wedge ... \wedge X_n$ is unsatisfiable. Hence, the formula $X$:

$$\sim \left(X_1 \wedge X_2 \wedge ... \wedge X_n\right)$$

is valid. Thus there is a regular set $R$ such that $X$ is tautologically implied by $R$, and such that no critical parameters of $R$ occur in $X$. Thus $R \bigcup \sim X$, that is:

$$R \bigcup \left(X_1 \wedge X_2 \wedge ... \wedge X_n\right)$$

is unsatisfiable, which is to say

$$R \bigcup S$$

is unsatisfiable.

(b): Next, suppose that every unsatisfiable set $S$ has an assosciate. We must show that every valid sentence $X$ is tautologically implied by some regular set $R$ such that no critical parameter of $R$ occurs in $X$. Well, suppose some sentence $X$ is valid. Thus, the set $\sim X$ is unsatisfiable, and it therefore has an assosciate $R$. Hence $R \bigcup \sim X$ is unsatisfiable and no critical parameters of $R$ are in $\sim X$. Thus $X$ is tautologically implied by $R$, and is such that no critical parameters of $R$ occur in $X$.

## Grand Exercise

$T_1$: We first show that $tableau - consistency$ is an analytic consistency property. To review, an analytic consistency property $\Gamma$ is one such that for any set $S$ having that property:
$\Gamma_0$: $S$ contains no element and its negation (or no element and its conjugate, if we are considering signed formulas).
$\Gamma_1$: For any $\alpha$ in $S$, the sets $S \bigcup \alpha_1$ and $S \bigcup \alpha_2$ are both $\Gamma$-consistent.
$\Gamma_2$: For any $\beta$ in $S$, at least one of the sets $S \bigcup \beta_1$ and $S \bigcup \beta_2$ is $\Gamma$-consistent.
$\Gamma_3$: For any $\gamma$ in $S$ and any parameter $a$, the set $S \bigcup \gamma(a)$ is $\Gamma$-consistent.
$\Gamma_4$: For any $\delta$ in $S$, the set $S \bigcup \delta(a)$ is $\Gamma$-consistent for any parameter $a$ that does not occur in any element of $S$.
So, let tableau-consistenty be the property $\Gamma$. $\Gamma_0$ is obvious, as there existing no closed tableau for $S$ by definition means $S$ cannot contain any formula $X$ and its conjugate. $\Gamma_1 - \Gamma - 4$ are also obvious, as a fully completed tableau for $S$ includes using rules $A, B, C, D$ as much as possible (and hence all the tableaus after using one extension by the rule must also be open, that is, the set of formulae on its branch must be tableau-consistent).

$T_2$: Next, we show that satisfiablity is an analytic consistency property. By the definition of satisfiability, $\Gamma_0$ is immediate. Now, if $S$ is satisfiable, then under any interpretation $I$ under which it is satisfiable, any $\alpha \in S$ is also true, and hence both $\alpha_1$ and $\alpha_2$ would be true, meaning $\Gamma_1$ holds. $\Gamma_2 - \Gamma_4$ hold in a similar fashion.

$T_3$: We need to show that $F$-consistency (that is, the property that all finite subsets of $S$ are satisfiable) for a set $S$ is an analytic consistency property. Suppose $\Gamma_0$ didn't hold. Thus $X, \sim X \in S$ for some $X$. But this contradicts the assumption that all finite subsets of $S$ were satisfiable. As for $\Gamma_1$, suppose that some $\alpha \in S$. Now, by the assumption that $S$ is $F$-consistent, all finite

subsets of $S$ are satisfiable, and hence all finite subsets of $S \bigcup \alpha_1$ that do not include $\alpha_1$ are satisfiable. Now, consider some finite subset of $S \bigcup \alpha_1$ of the form $\alpha_1, X_1, X_2, ..., X_n$ for elements $X_1, X_2, ..., X_n$ of $S$. Well, it was assumed that the set $X_1, X_2, ..., X_n$ was satisfiable (as it was in $S$). Also, because $\alpha \in S$, $X_1, X_2, ..., X_n, \alpha$ is satisfiable under some interpretation $I$. Because $\alpha_1$ would then also be true under $I$, $X_1, X_2, ..., X_n, \alpha_1$ is therefore true under $I$, and hence satisfiable. The case for $\alpha_2$ is identical, except replacing $\alpha_1$ with $\alpha_2$. $\Gamma_2 - \Gamma_4$ can be shown to hold in a similar fashion as in the $\alpha$ case.

$T_5$: Next, we show that $A$-consistency (that is, that a set $S$ does not have an assosciate) is an analytic consistency property. Now, suppose that $S$ is $A$-consistent. This means that $S$ must not be unsatisfiable (for if it were, $R \bigcup S$ would obviously be unsatisfiable for every regular set $R$). This of course means that $\Gamma_0$ is satisfied, as having some formula $X \in S$ and $\sim X \in S$ would make $S$ unsatisfiable. Next, as for $\Gamma_1$, suppose that some $\alpha \in S$. Now, because $S$ is $A$-consistent, $R \bigcup S$ is satisfiable for every regular set $R$ under some interpretation $I$. Hence, under $I$, $S$ would be simultaneously satisfiable, and thus $\alpha$ would be true, thus meaning both of $\alpha_1, \alpha_2$ are also true. Thus, $R \bigcup S, \alpha_1$ and $R \bigcup S, \alpha_2$ would be satisfiable for every regular set $R$ under the interpretation $I$ under which $R \bigcup S$ was satisfiable. $\Gamma_2 - \Gamma_4$ can be shown using a similar idea.

$T_6$: Lastly, we show that Craig-consistency (that is, a set $S$ in which there is a partition $S_1|S_2$ of $S$ such that there is no partition interpolant $X$ and $\sim X, X_1, X_2, ..., X_n$. Thus, this would mean $S$ is *not* Craig-consistent, and hence, by contradiction, $\Gamma_0$ holds. Now, as for $\Gamma_1$, suppose that some $\alpha \in S$. Consider the partition $S_1|S_2$ of $S$ such that there is no partition interpolant between $S_1$ and $S_2$. Without loss of generality, let $\alpha \in S_1$. Either $S_1 \bigcup \sim Z$ or $S_2 \bigcup Z$ is satisfiable for every sentence $Z$ such that all predicates and parameters of $Z$ occur in both $S_1$ and $S_2$. If the latter case holds for the given $Z$, we immediately know that the partition $\alpha_1, S_1|S_2$ of $S \bigcup \alpha_1$ would be such that $S_2 \bigcup Z$ is satisfiable. If it was the former for the given $Z$, then, under the interpretation $I$ under which $S_1 \bigcup \sim Z$ was satisfiable, $\alpha$ would be true, and hence so would $\alpha_1$, meaning the partition $\alpha_1, S_1|S_2$ of $S \bigcup \alpha_1$ would be such that $\alpha_1, S_1 \bigcup \sim Z$ would be satisfiable. In any case, either of $\alpha_1, S_2 \bigcup \sim Z$ or $S_2 \bigcup Z$ would be satisfiable for every $Z$, thus meaning $S \bigcup \alpha_1$ is Craig-consistent for the partition $\alpha_1, S_1|S_2$. The case for $\alpha_2$ is identical to the above case for $\alpha_1$. The cases for $\Gamma_2 - 4$ can be seen with a similar idea.

# 10

Suppose that $M$ is both maximally $\Gamma$-consistent and $E$-complete. We must show that it is a Hintikka set.

$H_0$: We must show that no sentence $X$ and its conjugate $\bar{X}$ are in $S$. This is immediate from condition $\Gamma_0$.

$H_1$: We must show that for any $\alpha \in S$, both $\alpha_1$ and $\alpha_2$ are in $S$. Well, because $M$ is $\Gamma$-consistent, both $M \bigcup \alpha_1$ and $M \bigcup \alpha_2$ are $\Gamma$-consistent. But suppose $\alpha_1$ ($\alpha_2$) was not in $M$. Thus $M$ would be the proper subset of $M \bigcup \alpha_1$ ($M \bigcup \alpha_2$), and $\Gamma$-consistent set, contrary to the assumption that $M$ was maximally $\Gamma$-consistent. Hence both $\alpha_1$ and $\alpha_2$ are in $S$.

$H_2$: We must show that for any $\beta \in S$, either $\beta_1$ or $\beta_2$ are in $S$. Because $M$ is $\Gamma$-consistent, either $M \bigcup \beta_1$ or $M \bigcup \beta_2$ are $\Gamma$-consistent. Without loss of generality, suppose it was $M \bigcup \beta_1$ which was $\Gamma$-consistent. Then $\beta_1 \in S$, as, if it weren't, $M$ would be a proper subset of the $\Gamma$-consistent set $M \bigcup \beta_1$, contrary to the assumption that $M$ was maximally $\Gamma$-consistent. Thus either $\beta_1 \in S$ or $\beta_2 \in S$.

$H_4$: We must show that, for any $\gamma \in S$, $\gamma(a) \in S$ for every parameter $a$. Because $M$ is $\Gamma$-consistent, $M \bigcup \gamma(a)$ is $\Gamma$-consistent for every parameter $a$. Hence, by the assumed $\Gamma$-consistency maximality of $M$, $\gamma(a) \in S$ for every parameter $a$.

$H_4$: We must show that, for any $\delta \in S$, there is at least one formula of the form $\delta(a) \in S$. This is immediate due to $M$ being $E$-complete.